

Algebra 1
Unit 14
Lesson H1 Practice

Name Answer Key
Date _____
Period _____

High school students in the United States were invited to complete an online survey in 2020. Part of the survey included questions about superhero powers. More than 1,000 students responded to this survey that included a question about a favorite superhero power. Researchers randomly selected 450 of the completed surveys. The two-way table broken down by gender the results.

	To Fly	Freeze Time	Invisibility	Super Strength	Telepathy	Total
Females	49	60	48	1	70	228
Males	51	71	27	25	48	222
Total	100	131	75	26	118	450

1. Do you have enough information to create a two-way relative frequency chart from the information provided? Explain.

Yes, because you have the frequencies and totals for each category.

2. Complete the two-way relative frequency table to represent the data.

	To Fly	Freeze Time	Invisibility	Super Strength	Telepathy	Total
Females	10.9%	13.3%	10.7%	0.2%	15.6%	50.7%
Males	11.3%	15.8%	6%	5.6%	10.6%	49.3%
Total	22.2%	29.1%	16.7%	5.8%	26.2%	100%

Use the completed table from question 2 for questions 3–10.

3. What is the sum of the marginal relative frequencies?

100%

4. What is the sum of the joint relative frequencies for high school males?

49.3%

5. What is the sum of the joint relative frequencies for high school students who choose super strength as a superhero power?

5.8%

6. What percentage of those who chose telepathy are female?

15.6%

7. What percentage of females chose to freeze time?

13.3%

8. The values in question 6 and 7 represent

- ☐ joint
- ☐ marginal
- ☒ conditional

relative frequency.

9. Identify an example of a marginal relative frequency from the table.

Answers will vary.
Total high school students who chose "to fly"

10. Identify an example of a conditional relative frequency from the table.

Answers will vary.
Males who chose "to fly"